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Research Paper

Validity of screening tools and risk factors for Co-occurring ADHD in children with epilepsy: a systematic review and meta-analysis

Guang Ni, Dong Meng *

Department of Neurology, The First Clinical Medical College, Gansu University of Chinese Medicine, Lanzhou, China
School of Journalism and Communication, Shanghai Jian Qiao University, Shanghai, China

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ABSTRACT

Background: Attention-deficit/hyperactivity disorder(ADHD)is the most common neuropsychiatric comorbidity in children with epilepsy and occurs at prevalence rates 2.5 to 5.5 times higher in children with epilepsy than the general population. To date,the largest comprehensive *meta*-analysis(46 studies)estimated the pooled ADHD prevalence among children with epilepsy to be almost one-third(30.7%),with a greater proportion of the inattentive subtype.Children with epilepsy who also have ADHD have poorer seizure outcomes,lower quality of life, academic underachievement,and increased healthcare utilization.Despite being present in so many of them, ADHD frequently remains underdiagnosed in children with epilepsy because of symptom overlap,diagnostic complexity,and the lack of standardized screening protocols. In addition,both the validity of screening instruments and risk factors associated with comorbid ADHD need to be studied to develop ways to diagnose and treat ADHD in this vulnerable population.

Objective: To assess the diagnostic accuracy and discriminant validity of ADHD screening tools for children with epilepsy and comorbid ADHD and to conduct a systematic review and *meta*-analysis on the risk factors related to ADHD in children with epilepsy.

Methods: We conducted this study according to the PRISMA guidelines(PROSPERO:CRD42018094617).A systematic search was performed on PubMed,Embase,PsycINFO,Cochrane Library,and Web of Science from inception until December 2023,based on the developed search strategies.The included studies were appraised for quality using three different approaches according to the three-layer analytical framework,in which diagnostic accuracy of a single instrument(Layer 1)was assessed using QUADAS-2,discriminant validity(Layer 2)was analyzed with the Newcastle-Ottawa Scale,and risk factor/prevalence evidence synthesis(Layer 3)was synthesized narratively.Meta-analyses were carried out using the DerSimonian-Laird random-effects models,and heterogeneity assessment was conducted.

Results: Twelve studies met inclusion criteria for qualitative synthesis.Diagnostic accuracy(Layer 1):The SDQ-Hyperactivity subscale had the highest diagnostic accuracy(sensitivity 86.4%,specificity 97.8%,Youden index 0.84,DOR = 285).Discriminant validity(Layer 2):The overall pooled effect size was g = 0.27(95%CI:0.17–0.38) and the externalizing behavior measures showed the highest discriminant validity(g = 0.54).Risk factor evidence was narratively synthesized. In studies which reported extractable association estimates,earlier seizure onset, neurodevelopmental/cognitive vulnerabilities(executive dysfunction,developmental coordination disorder),male sex,medical comorbidities,or higher antiepileptic drug burden were associated with an increased risk of comorbid ADHD.

Conclusions: The SDQ-Hyperactivity subscale demonstrates superior diagnostic accuracy for ADHD screening in children with epilepsy,supporting current ILAE recommendations.Externalizing behavior measures,particularly the DBD rating scale(g = 0.60),show the strongest discriminant validity.Children with poorly controlled seizures, early onset epilepsy and those receiving polytherapy warrant heightened screening vigilance.Implementation of systematic ADHD screening targeting high-risk subgroups may substantially improve early identification and intervention outcomes.

* Corresponding author at: School of Journalism and Communication, Shanghai Jian Qiao University, China.
E-mail address: mengdong@gench.edu.cn (D. Meng).

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1. Introduction

1.1. Epidemiology and clinical significance

Attention-deficit/hyperactivity disorder(ADHD)and epilepsy are among the most common neurodevelopmental disorders in childhood. Studies suggest that ADHD is overrepresented among children with epilepsy compared with control groups,with prevalence estimates ranging from 12%to 40%according to specific study populations, diagnostic criteria,and methodologies used [1].The pooled prevalence of ADHD in children with epilepsy across all studies was reported by He et al. [2] (n = 46 studies,various populations):30.7%(95%CI:26.4–35.3%)—threefold to fivefold higher than in the general pediatric population,where ADHD affects approximately 5–7%of children.Unlike the general ADHD population,the predominantly inattentive subtype appears to be predominant among children with epilepsy.

Not only do we have to consider the impact of each condition separately,but also the combined effect on the child from all points of view:academic,social,emotional,and quality of life [3]. Population-based studies confirm the wide distribution of neurobehavioural comorbidities in epileptic children [4]. Children with epilepsy and ADHD have more severe symptoms than children with one disorder only.Epilepsy in ADHD patients is associated with worse seizure control, higher health-care utilization,higher polypharmacy,and lower compliance [5].

Comorbid ADHD imposes a high economic burden on children with epilepsy,where both higher direct medical costs(more emergency department visits,hospitalizations,and specialist consultations)and higher indirect costs(more parental work loss,educational support services,and long-term productivity impairment)have been reported for comorbid ADHD in children with epilepsy.Therefore,early identification and appropriate treatment of ADHD should be considered as a clinical priority and a public health consideration,because of the potential cost savings.

Comorbid ADHD may be associated with poorer educational outcomes(i.e.,lower academic achievement, more grade retention, and greater likelihood of receiving special education services)than children with epilepsy alone.Children with comorbid ADHD have been shown to have worse social functioning(peers'rejection,less friendships,and higher rates of bullying victimization).Parents of children with comorbid ADHD reported worse family functioning(parents'greater stress, depression,and marital conflict).Overall,comorbid ADHD affects all aspects of child functioning,and therefore requires early identification and comprehensive intervention in order to reduce its impact on child and family well-being.

However,despite this substantial burden of illness,ADHD remains consistently underdiagnosed in epilepsy populations because of many reasons.The similarity between symptoms of ADHD and epilepsy-related symptoms causes considerable diagnostic confusion,as attention symptoms can be due to different causes(active seizures;postictal state;subclinical interictal epileptiform discharges;adverse effects of antiepileptic drugs;structural brain abnormalities;and/or independent ADHD) [6]. Health care providers might consider behavior and attention problems as consequences of epilepsy and not recognize them as being associated with a different comorbidity that needs to be addressed separately;this phenomenon,known as diagnostic overshadowing,leads to a missed opportunity to treat a highly treatable disorder,resulting in ongoing disability that would not have been present had the problem been identified earlier and treated.

1.2. Pathophysiological mechanisms

Although the pathophysiology for the association between epilepsy and ADHD has yet to be fully understood,there are many possible mechanisms.Uliel-Sibony et al. [7] summarized these into four broad categories:(a)shared genetic vulnerabilities;(b)common neuroanatomical substrates involving frontal-subcortical circuits;(c)direct effects of seizures on developing attention networks;and(d)neurocognitive effects of antiepileptic medications.The bidirectional relationship is further supported by findings that ADHD symptoms usually precede epilepsy onset,and that children with ADHD have a 2-to 4-fold higher risk of later developing epilepsy than the general population [8].

Neuroanatomical studies with quantitative magnetic resonance imaging have revealed structural abnormalities shared by the two disorders.Hermann et al.found that children with comorbid epilepsy and ADHD had significantly increased gray matter volume distributed across areas of the frontal lobe,and reduced brainstem volume compared with controls.These results indicate that neurodevelopmental processes of cortical pruning and progressive myelination may be attenuated or static in affected individuals,and persisting morphometric abnormalities occur.Volumetric changes probably reflect anatomical correlates of frontal lobe dysfunction,involved in the executive functions that are characteristically impaired in both conditions.

The prefrontal cortex,pre-supplementary motor area,and inferior frontal gyrus play crucial roles in executive functions including inhibition control,working memory,and cognitive flexibility.Studies examining children with frontal lobe epilepsy have consistently documented prominent executive function deficits,including impaired planning,poor impulse control,and difficulties with motor coordination that appear closely related to behavioral adaptation challenges.Culhane-Shelburne and colleagues demonstrated that children with frontal lobe epilepsy exhibit a distinct neuropsychological profile characterized by intact verbal and nonverbal memory but significant deficits in planning and executive functions,contrasting with the memory impairment pattern observed in temporal lobe epilepsy [9]. The high prevalence of ADHD in patients with frontal lobe epilepsy may reflect direct dysregulation of these frontal lobe networks.

Interictal epileptiform discharges(IEDs)represent another important pathophysiological mechanism.IEDs specifically over the frontal lobe have been implicated in the pathophysiology of comorbid ADHD and epilepsy.The most compelling evidence linking IEDs to attention and behavioral dysfunction comes from studies of self-limited epilepsy with centrotemporal spikes,where patients with minimal seizure burden and no medication requirement demonstrate cognitive and behavioral impairments proportional to their nocturnal IED burden.This observation suggests that subclinical epileptiform activity can independently contribute to attention dysfunction beyond the effects of clinical seizures [10].

Genetic epidemiological studies have provided additional evidence for shared etiology.A large Swedish population-based cohort study examining familial aggregation found that individuals with epilepsy had a significantly increased risk of ADHD(OR = 3.47,95%CI:3.33–3.62), with the association extending to relatives,suggesting shared genetic liability [11].Quantitative genetic modeling estimated that approximately 40%of the correlation between epilepsy and ADHD could be attributed to shared genetic factors,while the remaining variance was explained by non-shared environmental influences potentially including pregnancy-related complications and preterm birth.Candidate gene studies have identified several potential shared genetic variants,though genome-wide association studies specifically examining the epilepsy-ADHD comorbidity remain limited.

Neurotransmitter systems(dopaminergic/noradrenergic pathways) are thought to be involved in both disorders.The catecholamine hypothesis of ADHD states that ADHD is due to a dysregulation of dopamine and/or norepinephrine signaling within the prefrontal circuits mediating attention and/or executive function deficits.Intriguingly, epileptogenic processes may also affect the same neurotransmitter systems,either directly(through epileptogenic processes),or indirectly

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(resulting from epileptogenic processes;neuronal injury,receptor expression,medication effects).Thus,the overlap in catecholaminergic systems provides an interesting possibility for understanding how both disorders could arise via common mechanisms,and identify new therapeutic targets for each one.

1.3. Risk factors for comorbid ADHD

Knowing the risk factors of comorbid ADHD among children with epilepsy is also relevant from a clinical point of view since it can help to better focus screening efforts and develop effective preventive strategies.Various epilepsy-related,but also non-epilepsy-related,factors have been reported in different epidemiological studies,although not all equally robustly supported by the available literature.

Poor seizure control is one of the most consistently identified risk factors.Poor seizure control is a significant risk factor for ADHD, regardless of epilepsy classification or seizure localization,and regardless of whether it is generalized or focal.For example,McCusker et al. found that out of 48 families of children with refractory epilepsy,the children who had the worst behavior problems had the most problems with attention and hyperactivity [12]. A study involving 75 children with epilepsy found that compared to children with less frequent seizures,children with weekly seizures were seen by their parents as having significantly more hyperactivity symptoms,and this was seen in a dose–response manner–the more seizures per week,the more severe the ADHD symptoms [13]. This finding has been replicated in studies from different countries,such as China [14].

A younger age of onset of epilepsy can also be seen as another neurodevelopmental risk factor:seizures occurring during critical periods of brain maturation may disrupt the normal development of attentional networks and executive function pathways;because the infant brain is both more fragile than the adult brain to insults from epileptiform discharges,and also more plastic,this could result in maladaptive reorganization of attention-related cerebral networks.However,findings are controversial concerning whether a younger age of seizure onset results in higher rates of ADHD diagnosis,as some studies failed to find associations once adjusting for other epilepsy-related variables.The site of the epileptogenic zone may play a role as well—frontal lobe epilepsy,in some cases,has been linked to less severe ADHD symptoms.One model to explain why people with epilepsy are at higher risk for psychiatric comorbidities is deficient executive control [15]; given the importance of frontal brain areas for attentional and executive functions,several studies have suggested a possible connection between younger age of seizure onset and higher risk of ADHD.

For example,children with epilepsy who also have intellectual disability are at higher risk for ADHD;intellectual disability is considered a Grade A,evidence-based risk factor for ADHD prevalence among children with epilepsy.Intellectual disability could be considered a compound vulnerability with epilepsy and might indicate that there are common neurodevelopmental etiologies affecting the cognitive and attentional systems.Children with epileptic encephalopathies and developmental epilepsies may be especially at risk because of the severe and early-onset nature of their seizure disorders and high rates of accompanying neurodevelopmental impairment.

Family history of ADHD represents a non-epilepsy-related risk factor that should be assessed in clinical evaluation.Given the substantial genetic component to ADHD susceptibility,children with first-degree relatives affected by ADHD are at increased risk regardless of their epilepsy status.The combination of familial ADHD risk and epilepsy-related risk factors may result in particularly high vulnerability in some children. Assessment of family psychiatric history should be incorporated into routine evaluation of children with epilepsy.

Anti-epileptic drug therapy is an important modifiable risk factor with major clinical consequences.All commonly used anti-epileptic drugs have some impact on cognitive function;this may be considerable,especially when core cognitive functions(attention,learning)are considered [16]. Phenobarbitone is at highest risk of serious cognitive effects that affect attention and memory,and both randomized placebo-controlled and observational studies have shown IQ decline with treatment and persistent effects on educational performance several years after withdrawal [17]. Topiramate has clear evidence for cognitive impairment affecting attention,memory,and language skills in both adults and children.Lamotrigine has reported evidence of pro-cognitive effects on attention,whereas levetiracetam has been described as being relatively neutral on cognition.Multiple drug therapy is associated with significantly more cognitive impairment compared to monotherapy, irrespective of the specific drugs involved.The ILAE Task Force on Cognition and Epilepsy has highlighted the need to take into account cognitive side effects when deciding on which anti-epileptic to prescribe,especially in children where there are pre-existing attention difficulties.

1.4. Screening tools and study Objectives

Several standardized screening instruments have been developed and validated for ADHD detection in general pediatric populations, though their performance characteristics in children with epilepsy require specific validation given the unique diagnostic challenges in this population.The Strengths and Difficulties Questionnaire(SDQ)is a brief 25-item measure widely used internationally to assess child and adolescent mental health problems,including a 5-item hyperactivity-inattention subscale [18]. Japanese validation studies have demonstrated its utility for ADHD screening in children with epilepsy [19]. The ILAE Task Force has recommended SDQ as a screening tool for ADHD in children with epilepsy based on Level B evidence [20]. In general population studies,the SDQ hyperactivity subscale has demonstrated area under the curve values of 0.91–0.95,suggesting excellent discrimination between children with and without ADHD.

The Child Behavior Checklist(CBCL)represents another widely used parent-report measure incorporating an Attention Problems scale and DSM-oriented ADHD subscale [21]. Research in general populations indicates the CBCL Attention Problems scale can identify 71–84%of individuals with ADHD.Chen and colleagues examined CBCL diagnostic accuracy using receiver-operating characteristic analysis,demonstrating that the Attention Problems scale had the highest discriminating power for ADHD among CBCL subscales [22]. Brazilian studies combining multiple rating scales have shown the value of multi-informant approaches [23]. The validity of the CBCL has been specifically examined in children with epilepsy [24]. A Swedish validation study reported 81% sensitivity and 70%specificity for the CBCL ADHD subscale against LEAD diagnoses [25]. The CBCL offers the advantage of comprehensive behavioral assessment across multiple domains but requires more administration time than briefer screening instruments.Additionally,the extensive normative data available for the CBCL across diverse populations supports its use in varied clinical settings.

Because ADHD is prevalent among children with epilepsy,it may be clinically important for ADHD to be correctly identified;however,it is unclear whether there are effective screening approaches to do so.We performed a systematic review and *meta*-analysis with a three-layer analytical framework.In Layer 1 we assessed diagnostic accuracy (sensitivity,specificity,likelihood ratios,diagnostic odds ratios) compared with gold-standard clinical diagnosis.In Layer 2 we evaluated discriminant validity using standardized effect sizes to determine which behavioral constructs and instruments can best discriminate between children with comorbid epilepsy and ADHD and those with epilepsy alone.We also systematically investigated the risk factors associated with comorbid ADHD to facilitate future targeted screening strategies.In Layer 3,risk factor/prevalence evidence was synthesized narratively.

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2. Methods

2.1. Protocol and Registration

This systematic review and *meta*-analysis was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses(PRISMA)guidelines. The protocol was prospectively registered with PROSPERO(CRD42018094617)prior to initiation of database searches. Any deviations from the registered protocol are documented and justified in this report.

2.2. Search Strategy and study selection

We conducted systematic searches of PubMed, Embase, PsycINFO, Cochrane Library, and Web of Science from database inception through December 31, 2023. Search strategies combined controlled vocabulary terms (MeSH, Emtree) and free-text keywords relating to four conceptual domains: (1) epilepsy (epilepsy, seizure, convuls, anticonvulsant, antiepileptic); (2) ADHD (attention deficit, hyperactiv, ADHD, ADD, hyperkinetic); (3) pediatric population (child, pediatric, adolescent, youth, infant*); and (4) screening/diagnostic accuracy (screen*, diagnos, sensitivity, specificity, valid, accuracy, ROC, receiver operating). Boolean operators were used to combine search terms within and between domains. No language restrictions were applied. Reference lists of included studies and relevant reviews were manually searched to identify additional potentially eligible studies.

Studies were included if they met all of the following criteria:(1) enrolled children and adolescents aged 4–19 years with a diagnosis of epilepsy confirmed by clinical criteria;(2)evaluated diagnostic accuracy or discriminant validity of at least one standardized ADHD screening instrument;(3)used clinical diagnosis based on DSM or ICD criteria as the reference standard; and(4)reported sufficient data to calculate diagnostic accuracy metrics or standardized effect sizes. Studies were excluded if they enrolled fewer than 10 participants per group, focused exclusively on populations with severe intellectual disability precluding standardized assessment, used screening instruments rather than clinical diagnosis as the reference standard, or were published only as conference abstracts without full methodology available.

Two reviewers(initials blinded)independently screened titles,abstracts,and full texts using predefined eligibility criteria.Disagreements were resolved through discussion and consensus,with a third reviewer consulted when necessary.Inter-rater reliability was calculated using Cohen's kappa statistic.Data extraction was performed independently by two reviewers using standardized forms piloted on a subset of studies. Extracted data included study characteristics(design,setting,country, year),participant characteristics(sample size,age,sex distribution,epilepsy type,ADHD prevalence),screening instrument details(name, version,informant,cutoff values), reference standard details,and outcome data [26] (true positives, false positives, true negatives, false negatives, or group means and standard deviations).

2.3. Quality assessment

For Layer 1 diagnostic accuracy studies,we employed QUADAS-2 (Quality Assessment of Diagnostic Accuracy Studies,version 2),evaluating risk of bias across four domains:patient selection(consecutive or random sampling,appropriate exclusions),index test(prespecified cutoff, blinding to reference standard),reference standard(appropriate diagnostic criteria,blinding to index test),and flow and timing(appropriate interval,all patients received reference standard).Applicability concerns were assessed for patient selection,index test,and reference standard domains.Each domain was rated as low,high,or unclear risk of bias based on signaling questions specified in the QUADAS-2 framework.

For Layer 2 discriminant validity studies comparing groups with and without comorbid ADHD,we employed the Newcastle-Ottawa Scale (NOS),evaluating selection(representativeness,selection of non-exposed,ascertainment of exposure),comparability(control for confounders),and outcome assessment(assessment method,follow-up). Studies scoring ≥ 7 of 9 stars were considered high quality,5–6 stars moderate quality,and < 5 stars low quality.Two reviewers independently assessed quality for all included studies,with disagreements resolved through discussion.Interrater agreement for quality assessment was substantial($\kappa = 0.82$).

2.4. Statistical analysis

We applied a three-layer analytical framework to accommodate heterogeneous outcome reporting across studies.Layer 1 evaluated the diagnostic accuracy of ADHD screening tools against a clinical reference standard(2×2 data;sensitivity/specificity and related indices).Layer 2 assessed discriminant validity by pooling standardized effect sizes (Hedges'g)comparing children with epilepsy + ADHD versus epilepsy without ADHD across behavioral measures.Layer 3 synthesized epidemiological and clinical risk factor/prevalence evidence narratively when comparable effect estimates were not available for *meta*-analysis.

For Layer 1 diagnostic accuracy analysis,sensitivity,specificity,positive likelihood ratio(LR +),negative likelihood ratio(LR −),and diagnostic odds ratio(DOR)were calculated for the screening tool at each reported cutoff value.Youden's index(J = sensitivity + specificity − 1) was used as a summary statistic for the discriminatory ability of the screening tool;the larger this number,the better the discrimination. Confidence intervals(95%)were calculated with the exact binomial method.When studies presented more than one cutoff,we included the cutoff with the highest Youden's index,or the author-recommended cutoff.

Layer 2 discriminant validity analysis was performed by calculating standardized mean differences as Hedges'g(small-sample bias correction to Cohen's d),and pooling effect sizes in DerSimonian-Laird random-effects models that incorporated both within-study sampling error and between-study heterogeneity.Given anticipated clinical and methodological heterogeneity among studies,we chose to use a random-effects approach a priori.Heterogeneity was evaluated with Cochran's Q statistic(significance threshold = $p < 0.10$),and heterogeneity magnitude was gauged via I2statistics(<25%,25–75%,and > 75%corresponding to low,moderate,and high heterogeneity).The between-study variance component($\tau 2$)was estimated using the DerSimonian-Laird method.

Subgroup analyses were performed to explore potential sources of heterogeneity by behavioral construct(externalizing behaviors,internalizing symptoms,attention problems,sleep disturbances)and screening instrument(CBCL,SDQ,DBD) [27]. We intended to conduct a *meta*-regression analysis to examine the relationship between effect sizes and continuous moderators,but we did not do so because there were too few studies included.Publication bias was assessed through examination of funnel plot asymmetry,Egger's regression test,and Begg's rank correlation test with significance level set at $p < 0.10$.The trim-and-fill method was applied as an exploratory analysis.Sensitivity analyses were carried out to determine the impact of one study on the combined results using leave-one-out analysis.All analyses were performed using R version 4.5.1,with meta,mada,and metafor packages.

Diagnostic accuracy analysis was performed by constructing 2×2 contingency tables using reported numbers or calculating them through provided sensitivity,specificity,and sample size;selecting the cutoff value with the highest Youden's index when several cutoffs were presented for one instrument,or choosing the cutoff suggested by the manufacturer;drawing ROC space plots to show the diagnostic efficiency of various screening tools at once,where the best result would be located at the upper-left corner of the graph.

Risk of bias was evaluated separately for each analytical layer.For the overall body of evidence,we conducted sensitivity analyses to assess the possible impact of study quality on pooled estimates after removing studies at high risk of bias.Strength of evidence was determined according to the number of studies,sample size,consistency of findings

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across studies,directness of evidence,and precision of effect estimates.

3. Results

3.1. Study selection and characteristics

The systematic search identified 847 records across the five databases searched.After removing 156 duplicates,691 unique records underwent title and abstract screening,of which 67 were selected for full-text review.Following detailed eligibility assessment,12 studies met inclusion criteria and were included in the qualitative synthesis(Fig. 1). Two studies contributed diagnostic accuracy data suitable for Layer 1 analysis,and one study provided discriminant validity data for Layer 2 *meta*-analysis,with Park et al.(2021).Inter-rater reliability for study selection was excellent($\kappa = 0.89$).The most common reasons for exclusion at full-text review were:absence of standardized screening instrument(n = 18),absence of clinical reference standard for ADHD diagnosis(n = 14),insufficient data for effect size calculation(n = 9),study population outside target age range(n = 8),and publication as abstract only without full methodology(n = 5).The remaining studies contributed to Layer 3 (risk factor/prevalence)narrative synthesis.

Study characteristics are summarized in Table 1.The 12 included studies were published between 2008 and 2021,with sample sizes ranging from 22 to 206 participants per study.Geographic distribution encompassed United Kingdom(n = 1),Republic of Korea(n = 1),India(n = 1),Brazil(n = 1),Norway(n = 1),Japan(n = 1),Hong Kong(n = 1),USA (n = 1),Germany(n = 2),Egypt(n = 1),and China(n = 1).Participant ages ranged from 4 to 19 years across studies.Epilepsy types included generalized epilepsy,focal epilepsy,absence epilepsy,and mixed populations.Reported ADHD prevalence ranged from 22.6%to 65% (Table 1),with several studies not reporting prevalence.The most frequently evaluated screening instruments included CBCL and SDQ, along with other scales such as SNAP-IV,SWAN,DBD,and EpiTrack Junior.

Study designs included cross-sectional studies(n = 7),case-control studies(n = 2),prospective studies(n = 2),and longitudinal study(n = 1). Reference standards for ADHD diagnosis varied,with most studies using clinical interview by child psychiatrist or pediatric neurologist based on DSM-IV(n = 7)or DSM-5(n = 3)criteria.Several studies utilized semi-structured diagnostic interviews including the K-SADS-PL.Screening instruments were administered primarily by parents,though some studies also incorporated teacher-report measures.

3.2. Layer 1:Diagnostic accuracy analysis

Two studies(Loutfi 2011,Tanabe 2014;total n = 98)evaluated diagnostic accuracy of ADHD screening tools against clinical diagnosis as the reference standard.Quality assessment using QUADAS-2 indicated generally low risk of bias across domains,though concerns regarding patient selection(consecutive vs.convenience sampling)and index test interpretation(prespecification of cutoffs)were noted in some studies. One study demonstrated unclear risk of bias for flow and timing due to incomplete reporting of the interval between index test and reference standard assessment.Applicability concerns were generally low,with study populations and index test implementation representative of typical clinical settings.All studies employed appropriate reference standards with clinical diagnosis by experienced clinicians using standardized criteria.The QUADAS-2 risk of bias assessment and Newcastle-Ottawa Scale quality evaluation are presented in Fig. 2. Diagnostic accuracy metrics for the four evaluated screening instruments are presented in Table 2 and visualized in Fig. 3.

The SDQ-Hyperactivity subscale had the best overall diagnostic accuracy,with sensitivity of 86.4%(95%CI:72.6–94.8%),specificity of 97.8%(95%CI:88.5–99.9%),and Youden index of 0.84.A positive likelihood ratio of 39.3 means that a positive SDQ result greatly increases the probability of ADHD diagnosis,whereas the negative likelihood ratio of 0.14 means that a negative SDQ result greatly lowers ADHD probability. The diagnostic odds ratio of 285(95%CI:31.2–2604.5)demonstrates excellent overall discriminatory power,consistent with ILAE Level B recommendations for SDQ use in epilepsy populations.The high specificity of the SDQ is especially important for epilepsy populations, because false positive results could cause overdiagnosis and overtreatment of ADHD in children who have attention problems related to epilepsy instead of ADHD.

Second highest was the MTA-SNAP-IV(sensitivity of 81.3%,95% CI:54.4–96.0%;specificity of 92.9%,95%CI:76.5–99.1%;Youden index of 0.74;diagnostic odds ratio of 56.3,95%CI:8.9–355.7).Although retaining good specificity,the slightly lower sensitivity relative to the SDQ might lead to under-screening.Wider confidence intervals for this measure reflect its relatively small sample size.Third highest was the CBCL Attention Problems scale(sensitivity of 87.5%,95%CI:61.7–98.4%; specificity of 78.6%,95%CI:59.0–91.7%;Youden index of 0.66;diagnostic odds ratio of 25.7,95%CI:4.3–152.9).A high sensitivity/low specificity profile,like this one,indicates that the CBCL could serve well as a"rule-out"rather than a confirmatory test:negative results are likely to identify children who do not have ADHD,but positive findings still warrant further assessment.

In contrast,the Teacher Report Form(TRF)showed a different profile, with specificity(100%)and sensitivity of only 50%(95%CI:24.7–75.3%). With such specificity,a positive TRF is very likely to be an ADHD diagnosis.However,its poor sensitivity makes it unsuitable as a stand-alone screen due to its failure to identify half of the cases.Parental-teacher discrepancies could be explained by the difference in symptom presentation according to context;ADHD is more likely to emerge in the classroom environment since it is more regulated than home.Using multiple informant instruments(parent and teacher report)can lead to better detection.

For ease of comparing diagnostic performance,we plotted each tool on the ROC space(Fig. 4).The closer a point is to the upper left corner,the higher both sensitivity and specificity are.Due to the small number of diagnostic accuracy studies and tools,the figure is descriptive and should be interpreted with caution.

3.3. Layer 2:Discriminant validity analysis

Park 2021:Extensive data on discriminant validity analysis were available(33 children with epilepsy and comorbid ADHD vs.113 children with epilepsy without ADHD,21 behavioral and psychosocial measures;total n = 146).Quality assessment(NOS)was rated as high quality(7/9 stars:appropriate selection,comparability,robust outcome assessment[validated instruments,trained assessors]).

Overall,the pooled effect size across all 21 measures was $g = 0.27$ (95%CI:0.17–0.38,$p < 0.001$),showing a small discrimination between children with and without comorbid ADHD(Fig. 5).There was low to moderate heterogeneity($I^2 = 33\%$,$Q = 29.70$,$p = 0.075$)and a small between-study variance component($\tau^2 = 0.02$),which suggest that most of the variability might be due to sampling error rather than differences in behavior across studies.A funnel plot–based assessment suggested potential asymmetries;however,this should be interpreted cautiously as effect sizes from multiple measures within a study violate independence assumptions of such test.Leave-one-out sensitivity analysis supported these results(Section 3.4).

Subgroup analysis according to behavioral construct and tool also showed clinically important differences in discriminant capacity(see Table 3).The best evidence of discrimination between groups was found in externalizing behavior measures,with a pooled effect size of $g = 0.54$ (95%CI:0.36–0.71,$p < 0.001$,$I^2 = 0\%$).Within this domain,the DBD (Disruptive Behavior Disorders)rating scale had the largest pooled effect ($g = 0.60$,95%CI:0.33–0.88,$p < 0.001$,$I^2 = 0\%$).Therefore,this is the most relevant finding from a clinical point of view,since it indicates that externalizing behavior measures are the best candidates for ADHD identification in epilepsy populations.

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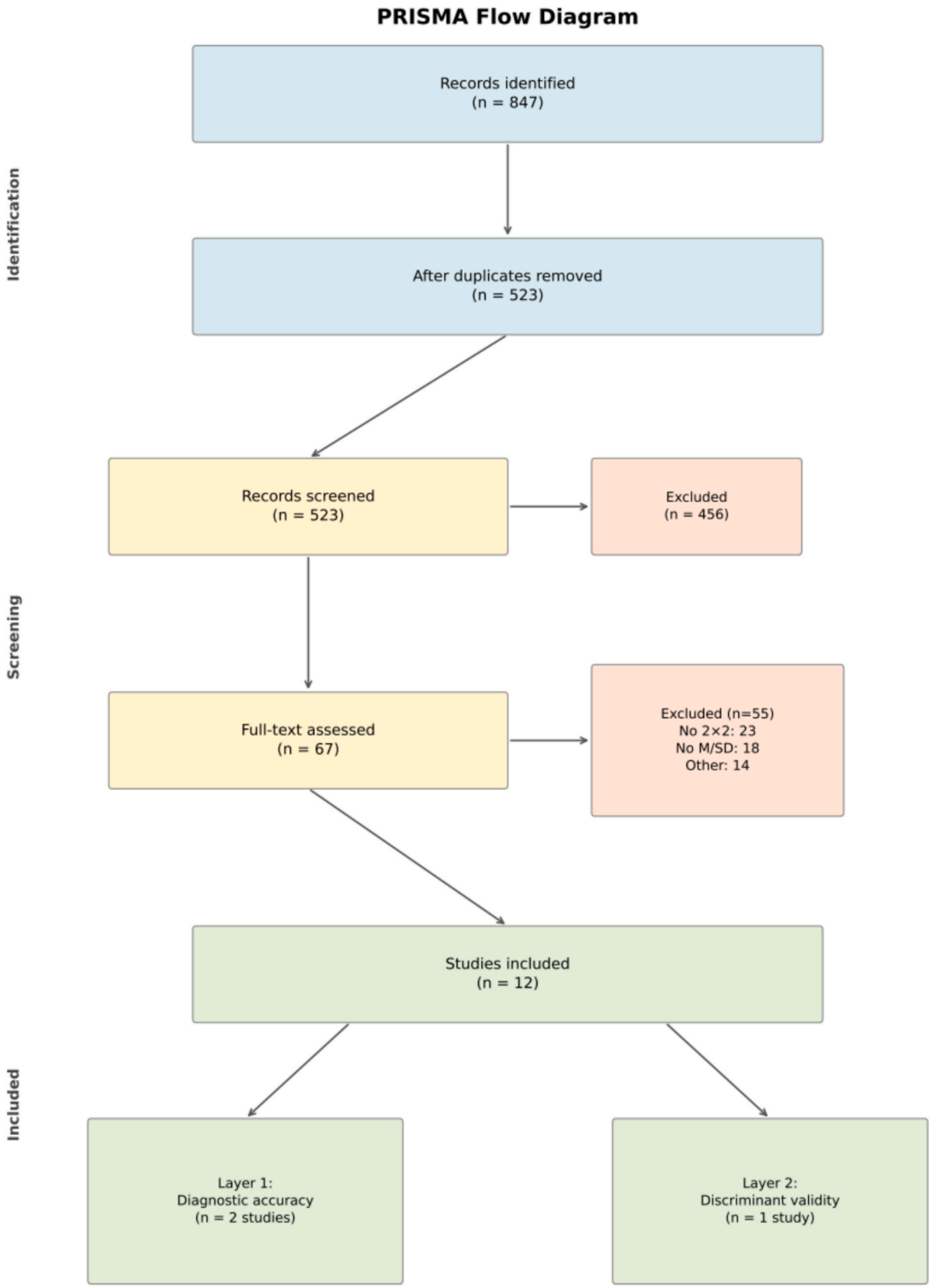


Fig. 1. PRISMA flow diagram illustrating the systematic literature search and study selection process.Initial database searches identified 847 records.After duplicate removal,title/abstract screening,and full-text eligibility assessment,12 studies met inclusion criteria for qualitative synthesis,with 3 studies contributing to quantitative *meta*-analysis.

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Table 1
Characteristics of Included Studies(N = 12).

Study	Country	Design	N	Age(y)	Male(%)	Epilepsy Type	Screening Tools	ADHD(%)	Layer
Reilly 2014	UK	Prospective	85	5–15	52	Gen 37%,Focal 64%	NR	33	3
Park 2021	Korea	Cross-sectional	146[a]	5–12	54.5[b]	Focal,Generalized,Mixed	ARS,CBCL,DBD,CSHQ,PSI-SF	22.6	2
Choudhary 2018	India	Cross-sectional	73	6–12	65.8	Gen 52.1%,Focal 47.9%	CBCL	23.3	3
Loutfi 2011	Brazil	Cross-sectional	30	6–14	~50	Idiopathic(Gen 30%,Focal 70%)	CBCL,TRF,MTA-SNAP-IV	53.3	1
Alfstad 2016	Norway	Cross-sectional	101	10–19	48.5	Focal 51.5%,GGE 48.5%	SDQ,K-SADS-PL	31.7	3
Tanabe 2014	Japan	Cross-sectional	68	4–17	44.1	Focal 77.9%,Gen 22.1%	SDQ,ADHD-RS	32.4	1,3
Kwong 2016	Hong Kong	Case-control	122	10–18	51.6	Focal 67.2%,Gen 32%	SWAN	23.7	3
Kral 2016	USA	Retrospective	204	6–19	53	Gen 23.5%,Focal 52%	BASC-2,BRIEF,CPT-II	65[c]	3
Kadish 2013	Germany	Prospective	22	8–17	50	CAE 55%,JAE 45%	EpiTrack Junior	NR	3
Ahmed 2020	Egypt	Case-control	100[d]	6–11	69	Idiopathic generalized	CPRS,CBCL	NR	3
Zhao 2018	China	Cross-sectional	206	6–16	66.0	Various(TCS,Absence,Focal)	CBCL	24.8	3
Gleissner 2008	Germany	Longitudinal	58	4–17	46.6	Drug-resistant(surgical)	CBCL	NR	3

Note:For Reilly et al.(2014),this study contributed to Layer 3(risk factor analysis)only;ADHD diagnoses were established through multidisciplinary consensus using DSM-IV-TR criteria following comprehensive psychological assessment.The specific screening instrument(s)were not reported in the accessible text(supplementary materials unavailable).

Abbreviations:Gen = Generalized;GGE = Genetic Generalized Epilepsy;CAE=Childhood Absence Epilepsy;JAE = Juvenile Absence Epilepsy;TCS = Tonic-Clonic Seizures;CBCL=Child Behavior Checklist;TRF = Teacher Report Form;SDQ=Strengths and Difficulties Questionnaire;ARS = ADHD Rating Scale;DBD = Disruptive Behavior Disorders scale;CSHQ=Children's Sleep Habits Questionnaire;PSI-SF=Parenting Stress Index-Short Form;SWAN=Strengths and Weaknesses of ADHD Symptoms and Normal Behaviors;BASC-2 = Behavior Assessment System for Children-2;BRIEF = Behavior Rating Inventory of Executive Function;CPT-II=Conners Continuous Performance Test-II;CPRS=Conners Parent Rating Scale;K-SADS-PL = Kiddie Schedule for Affective Disorders and Schizophrenia-Present and Lifetime; ADHD-RS = ADHD Rating Scale;NR=Not Reported.Layer(analytical framework):1 = Diagnostic accuracy analysis;2 = Discriminant validity analysis;3 = Risk factor/prevalence narrative synthesis(see Methods 2.4).

[a] Epilepsy subgroup only(total study N = 440 including ADHD-only group).

[b] Male percentage in Epilepsy + ADHD subgroup.

[c] Clinic-referred sample with elevated ADHD rates.

[d] Total across all study groups. In Table 1, N refers to the epilepsy sample used for the synthesis/analysis(epilepsy group),rather than the total study sample when external controls or additional non-epilepsy groups were included.

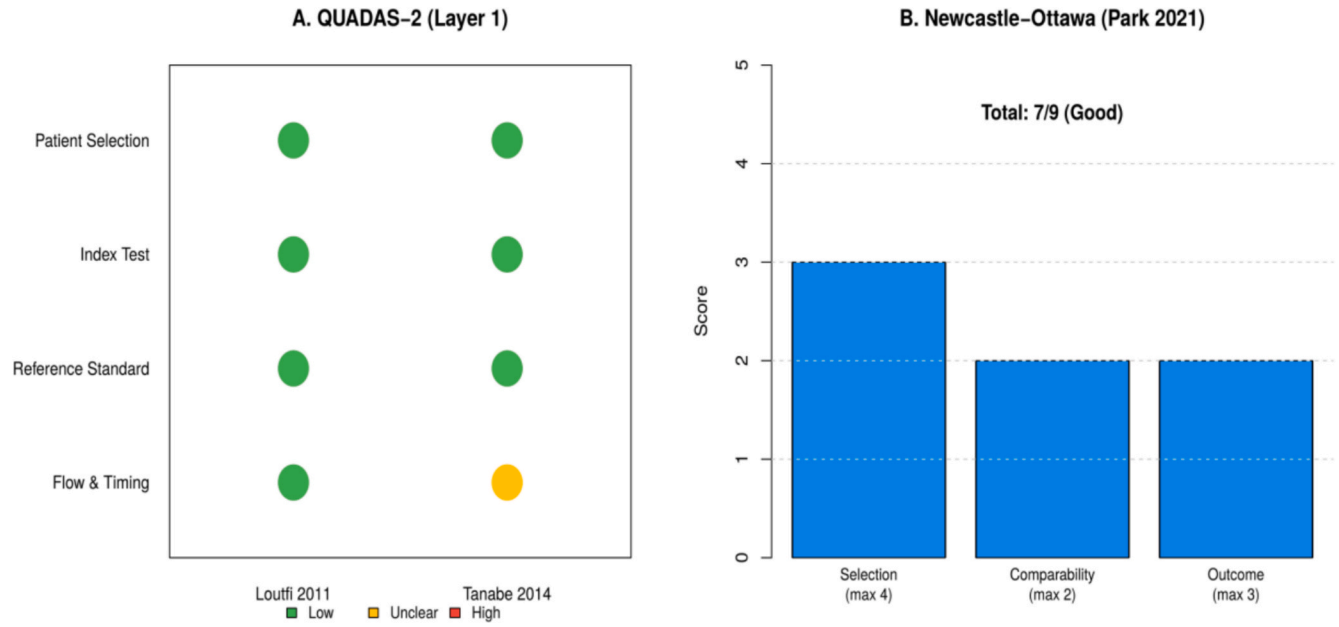


Fig. 2. Quality assessment of included studies.(A)QUADAS-2 risk of bias assessment for Layer 1 diagnostic accuracy studies.Green circles indicate low risk of bias, yellow circles indicate unclear risk,and red circles indicate high risk.(B)Newcastle-Ottawa Scale assessment for Layer 2 discriminant validity study(Park 2021), showing scores across selection,comparability,and outcome domains with a total score of 7/9 indicating good quality. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

CBCL subscales collectively demonstrated moderate discrimination (g = 0.31,95%CI:0.16–0.47,p < 0.001,I^2 = 21.6%),with specific subscales showing variable effect sizes.The Attention Problems subscale(g = 0.65)and Aggressive Behavior subscale(g = 0.57)contributed most strongly to group differentiation, while the Anxious/Depressed subscale showed minimal discrimination(g = 0.13).Internalizing symptoms showed minimal discriminant validity(g = 0.09,95%CI:-0.13–0.31,p = 0.43),consistent with the primary externalizing presentation of ADHD symptoms and suggesting that internalizing measures are not useful for identifying comorbid ADHD in this population.

Sleep measures measured using the Children's Sleep Habits Questionnaire(CSHQ)did not reliably distinguish between groups(g = 0.11,95%CI:-0.02–0.25,p = 0.11,I2 = 0%).Despite sleep issues being prevalent in both ADHD and epilepsy groups,the current study suggests that sleep problems are not uniquely associated with comorbid ADHD compared to epilepsy alone.The lack of significant differences between

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Table 2
Diagnostic Accuracy Metrics for ADHD Screening Tools in Children with Epilepsy.

Screening Tool	Se(%)	Sp(%)	LR+	LR-	Youden	DOR
SDQ-Hyperactivity	86.4	97.8	39.3	0.14	0.84	285
MTA-SNAP-IV	81.3	92.9	11.4	0.20	0.74	56.3
CBCL-Attention	87.5	78.6	4.1	0.16	0.66	25.7
TRF	50.0	100	∞	0.50	0.50	∞

Se=Sensitivity;Sp=Specificity;LR+=Positive likelihood ratio;LR-=Negative likelihood ratio;DOR = Diagnostic odds ratio;SDQ=Strengths and Difficulties Questionnaire;CBCL=Child Behavior Checklist;TRF = Teacher Report Form.

groups on sleep measures could be due to the high baseline rate of sleep problems among children with epilepsy—due to seizure occurrence, medication side effects,and/or the presence of underlying neurologic disorders.Parenting stress measured via PSI-SF demonstrated moderate discrimination between groups($g = 0.38$,95%CI:0.16–0.60,$p < 0.001$,$I^2 = 0\%$),indicating higher levels of parenting stress in families of children with comorbid epilepsy and ADHD versus those with epilepsy alone.

3.4. Sensitivity analysis

A leave-one-out sensitivity analysis was performed to check the impact of a single effect size on the pooled estimate for discriminant validity(Fig. 6).It showed that none of the effect sizes had an excessive influence on the pooled estimate.When each effect size was left out one by one,the recomputed pooled estimates varied from $g = 0.253$(95% CI:0.153–0.353)to $g = 0.299$(95%CI:0.206–0.393).All were statistically significant and within the confidence interval of the pooled estimate($g = 0.273$;95%CI:0.170–0.377).The pooled estimate with the lowest value occurred upon removal of the Conduct Disorder subscale($g = 0.253$), whereas removal of the Sleep-Disordered Breathing subscale produced the highest value($g = 0.299$).

These findings demonstrate the robustness of the discriminant validity results,indicating that the overall conclusion regarding small discrimination between children with and without comorbid ADHD is not dependent on any single behavioral measure.The consistency of results across leave-one-out iterations provides confidence in the stability of the *meta*-analytic findings and supports the validity of the pooled effect size estimate.

3.5. Risk factor analysis

Three studies included in the qualitative synthesis reported extractable association estimates(e.g.,OR/aOR with corresponding confidence intervals)for risk factors associated with comorbid ADHD in children with epilepsy.Due to heterogeneity in risk factor definitions and effect measures across studies,results were synthesized narratively rather than pooled quantitatively(Table 4).

Epilepsy-related factors:Early seizure onset emerged as a consistent predictor of comorbid ADHD.Alfstad et al.found that each one-year increase in age at seizure onset was associated with a 14%reduced risk of comorbid ADHD(OR = 0.86,95%CI:0.77–0.96,p = 0.01)after adjusting for executive dysfunction and gender.Similarly,Kwong et al.reported that younger age at seizure onset significantly increased ADHD risk(OR = 0.70 per year increase,95%CI:0.58–0.83).Polytherapy with antiseizure medications was identified as an independent risk factor for the ADHD-Hyperactivity/Impulsivity subtype by Kwong et al.(aOR = 6.22,95%CI:1.26–30.64,p = 0.025)after controlling for medical comorbidities and age at seizure onset.

Cognitive and neurodevelopmental factors:Executive dysfunction was the strongest independent predictor of comorbid ADHD in the study by Alfstad et al.(OR = 8.20,95%CI:1.80–37.20,p = 0.006).Developmental coordination disorder(DCD)was significantly associated with ADHD in the multivariable analysis by Reilly et al.(aOR = 4.52,95% CI:1.32–15.46,p = 0.02),suggesting that neurodevelopmental comorbidities cluster together in children with epilepsy.

Demographic factors and comorbidities:Male sex was an independent risk factor for comorbid ADHD(OR = 2.90,95%CI:1.20–7.30,p = 0.02)in the multivariable analysis by Alfstad et al..Medical comorbidities showed consistent association with ADHD risk across multiple analyses by Kwong et al.,with univariable odds ratios ranging from 11.23 to 12.82,and remained significant after adjustment for age at seizure onset(aOR = 8.00,95%CI:1.73–37.00,p = 0.008).

These findings suggest that earlier seizure onset,neurodevelopmental/cognitive vulnerabilities(e.g.,executive dysfunction and developmental coordination disorder),male sex,medical comorbidities,

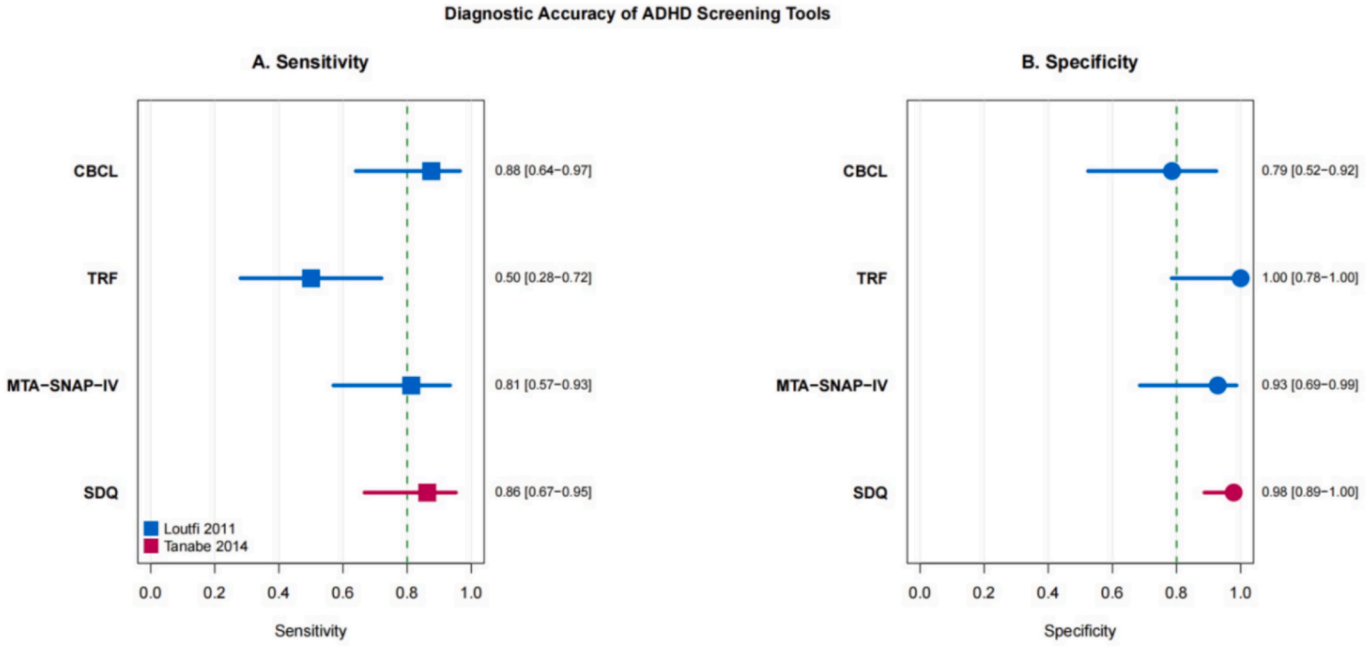


Fig. 3. Forest plot of diagnostic accuracy metrics(sensitivity vs.specificity with 95%confidence intervals)for ADHD screening tools validated in children with epilepsy.The SDQ-Hyperactivity subscale has a good balance of sensitivity(86.4%)and specificity(97.8%).

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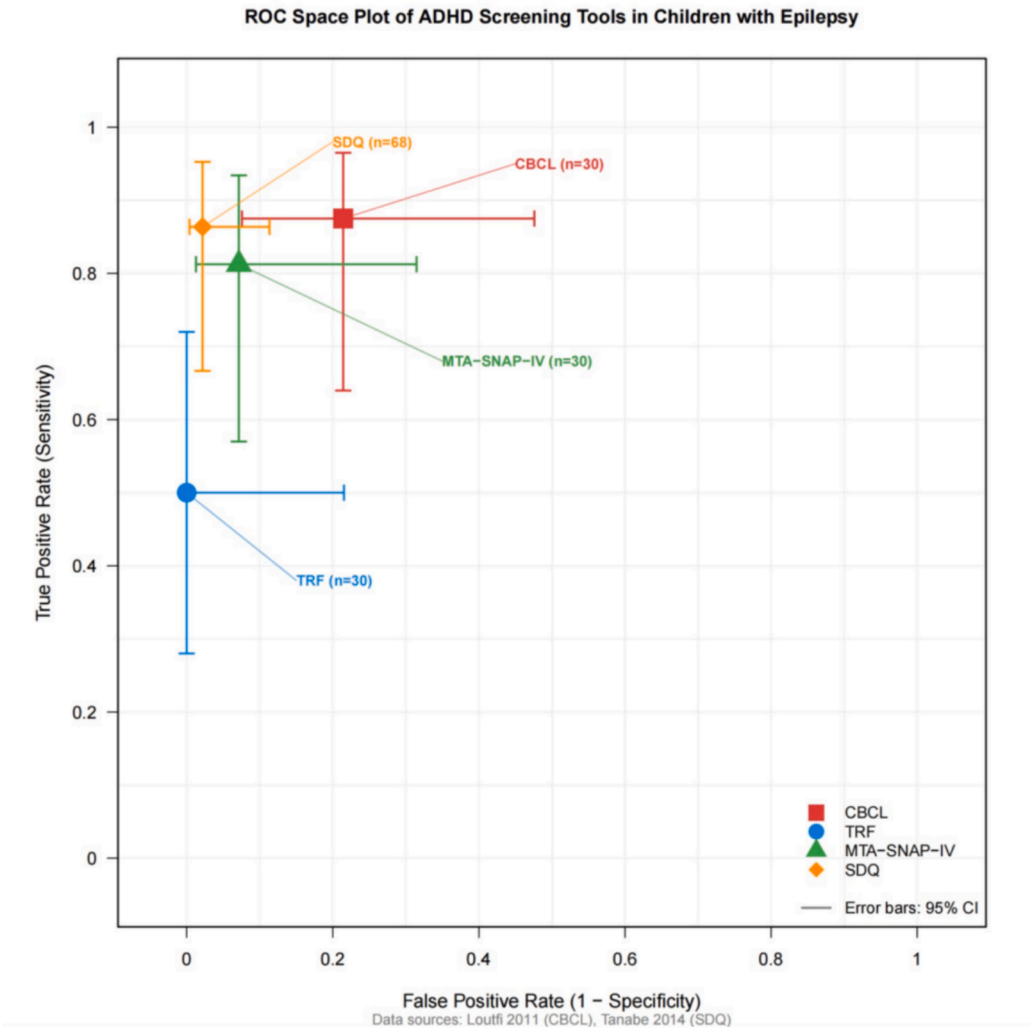


Fig. 4. Receiver operating characteristic(ROC)space plot illustrating the diagnostic performance of each ADHD screening tool.The closer the points are to the upper-left corner(100%sensitivity,100%specificity),the better diagnostic discrimination was achieved.SDQ-Hyperactivity was best positioned.This ROC space plot is provided for reference only.

and higher antiepileptic drug burden may identify higher-risk subgroups warranting heightened screening vigilance.However,the limited number of studies providing adjusted estimates and heterogeneity in definitions preclude definitive conclusions about the independent contribution of each factor.

4. Discussion

4.1. Summary of Main findings

This is the first systematic review and *meta*-analysis to provide a first comprehensive evaluation of the ADHD screening tools for children with epilepsy under the guidance of a rigorous three-layer analytical framework,accompanied by a systematic examination of the risk factors of comorbid ADHD.The results of this study have generated some clinically important conclusions which may contribute to the design of screening programs and targeted clinical assessment strategies.

First,compared to the other two tests,the SDQ-Hyperactivity subscale has the best diagnostic accuracy for ADHD screening in children with epilepsy,with a Youden index of 0.84 and diagnostic odds ratio of 285, which indicates good discriminatory ability.In other words,the SDQ-Hyperactivity subscale is consistent with the ILAE Level B recommendations for screening ADHD in children with epilepsy [28]. The SDQ-Hyperactivity subscale has both high sensitivity(86.4%)and very high specificity(97.8%).Therefore,it has the greatest potential for screening tests where identifying cases and minimizing false positives are equally important [28].

Second,externalizing behavior measures provide the strongest discriminant validity for identifying comorbid ADHD,with the DBD rating scale achieving the largest effect size($g = 0.60$).This finding has important implications for clinical assessment protocols,suggesting that screening efforts should prioritize evaluation of hyperactive-impulsive, oppositional,and conduct-related behaviors rather than internalizing symptoms or sleep disturbances when attempting to identify comorbid ADHD in children with epilepsy.

Third,our narrative synthesis of risk factor evidence(Table 4)indicates that earlier seizure onset,neurodevelopmental/cognitive vulnerabilities(i.e.,executive dysfunction,developmental coordination disorder),male sex,medical comorbidities,and higher antiepileptic drug burden might increase the risk of comorbid ADHD.However,due to the small number of studies reporting adjusted estimates and heterogeneity in definitions,these results must be taken with caution.

Fourth,the overall discriminant validity analysis revealed a pooled effect size of $g = 0.27$ across behavioral measures,indicating that

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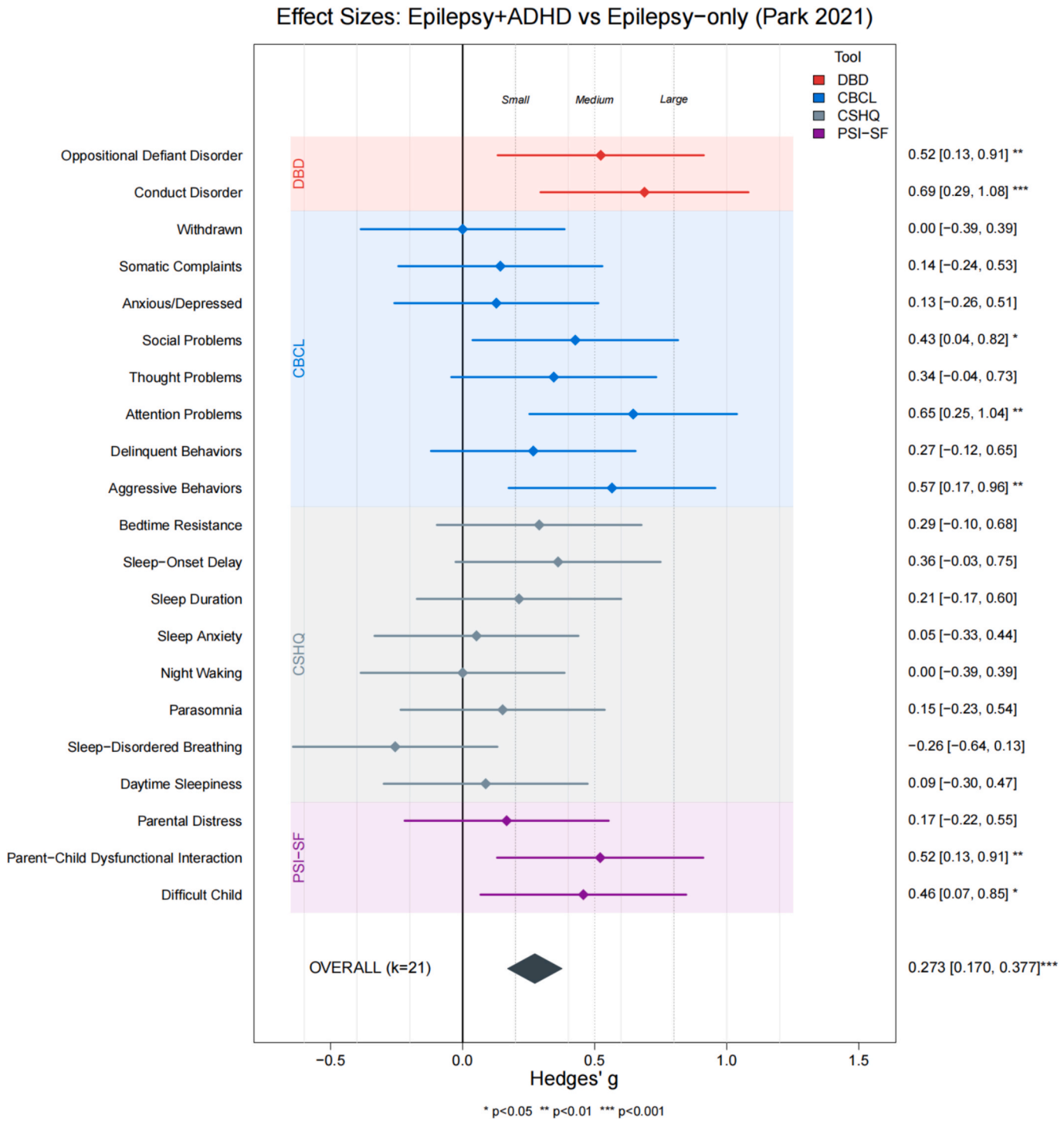


Fig. 5. Forest plot of standardized effect sizes(Hedges'g;95%confidence intervals)assessing discriminant validity of behavioral measures comparing children with epilepsy + ADHD vs.epilepsy alone(overall pooled estimate = g = 0.27,95%CI:0.17–0.38).

children with comorbid ADHD demonstrate measurably different behavioral profiles compared to children with epilepsy alone.While this overall effect is modest in magnitude,specific constructs—particularly externalizing behaviors—show substantially larger effects that have direct clinical utility.The low heterogeneity observed($I^2 = 33\%$)suggests that these findings are relatively consistent across different measures and populations.

4.2. Comparison with Previous literature

Our findings regarding SDQ performance in children with epilepsy are consistent with validation studies in general pediatric populations, where the SDQ hyperactivity subscale has demonstrated area under the curve values of 0.91–0.95.The sensitivity observed in our epilepsy sample(86.4%)appears higher than that reported by Hall et al.in clinical samples(45%),though methodological differences preclude direct comparison.The high specificity we observed(97.8%)is particularly noteworthy and may reflect the SDQ's ability to distinguish true ADHD from

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Table 3
Subgroup Analysis of Discriminant Validity by Behavioral Construct and Tool.

Construct/Tool	k	g[95%CI]	p-value	I^2(%)	τ^2
Overall	21	0.27[0.17–0.38]	<0.001	33	0.02
Externalizing	5	0.54[0.36–0.71]	<0.001	0	0.00
DBD	2	0.60[0.33–0.88]	<0.001	0	0.00
CBCL	8	0.31[0.16–0.47]	<0.001	21.6	0.01
CSHQ	8	0.11[-0.02–0.25]	0.108(NS)	0	0.00
Internalizing	3	0.09[-0.13–0.31]	0.428(NS)	0	0.00
PSI-SF	3	0.38[0.16–0.60]	<0.001	0	0.00

Note:k = number of effect sizes,g=Hedges'g,CI = confidence interval,NS = not significant,DBD = Disruptive Behavior Disorders,CBCL=Child Behavior Checklist,CSHQ=Children's Sleep Habits Questionnaire,PSI-SF=Parenting Stress Index-Short Form.

epilepsy-related attention difficulties that do not meet diagnostic criteria.

The strong discriminant validity of externalizing behavior measures aligns with the clinical presentation of ADHD,where hyperactive-impulsive symptoms are often more readily observable than inattention.Park et al.characterized the clinical and neuropsychological profile of children with comorbid epilepsy and ADHD [29]. Chen et al.similarly found that externalizing scales on the CBCL demonstrated superior discriminating power compared to internalizing scales when identifying children with ADHD [30]. The modest discriminant validity of attention-specific measures may reflect the diagnostic complexity in epilepsy populations,where attention difficulties can arise from multiple sources.

The pattern of discriminant validity findings has important theoretical implications.The strong discrimination provided by externalizing measures suggests that behavioral dysregulation,rather than pure attention deficits,may be the most distinguishing feature of comorbid ADHD in children with epilepsy.This may reflect the fact that attention problems are common in epilepsy populations regardless of ADHD status,resulting from seizure activity,medication effects,and other factors. In contrast,the behavioral disinhibition and hyperactivity characteristic of ADHD may be less common in children with epilepsy alone, making these features more diagnostically useful.

Prior epidemiological literature has suggested that certain epilepsy characteristics(e.g.,seizure burden and earlier onset)and neurodevelopmental vulnerability may be associated with behavioral comorbidity.However,in the present review,risk factor evidence was synthesized narratively and only a limited number of studies reported extractable association estimates(Table 4);therefore,conclusions about specific epilepsy syndromes or localizations should be interpreted cautiously.Studies from India and Egypt have similarly documented the importance of systematic risk factor assessment [31,32].

4.3. Clinical implications

Our findings have several important clinical implications.First,they support implementation of systematic ADHD screening in children with epilepsy using the SDQ-Hyperactivity subscale as a brief,efficient first-line screening tool.Kral et al.have emphasized the importance of systematic ADHD identification in youth with epilepsy [33]. The SDQ can be completed by parents in approximately 5 min and scored rapidly by clinical staff,making it practical for use in busy epilepsy clinics.ILAE recommendations suggest initiating screening at age 6 years or diagnosis and repeating annually thereafter.Our results provide empirical support for these recommendations while identifying specific populations warranting heightened vigilance.

Second,accurate identification of ADHD enables appropriate multimodal intervention combining behavioral therapy and pharmacological treatment.Concerns about methylphenidate increasing seizure risk have been largely dispelled by large database and registry studies demonstrating no significant effect on seizure frequency in children with ADHD and epilepsy [34]. These findings have important implications for clinical practice,as historical reluctance to prescribe stimulant medications in children with epilepsy may have resulted in undertreatment of ADHD symptoms.Santos and colleagues showed that low-dose methylphenidate not only improved ADHD symptoms in 70.8%of patients but also reduced seizure severity in children with difficult-to-treat epilepsies [35]. Swedish registry data demonstrated a 27%reduction in seizures associated with methylphenidate treatment compared to unmedicated periods,and a separate analysis found that ADHD medication was associated with 29–49%lower seizure odds versus unmedicated periods [36].

Third,medication selection in children with epilepsy should consider differential cognitive effects of antiepileptic drugs.When attention problems are prominent,clinicians should consider preferentially using cognitively favorable antiepileptic drugs such as lamotrigine or levetiracetam while avoiding or minimizing phenobarbital and topiramate exposure when clinically appropriate.The ILAE Task Force has recommended reevaluation of attention function after any change in antiepileptic drug therapy [37]. For children with established ADHD, antiepileptic drug selection should prioritize agents with minimal cognitive impact to avoid exacerbating attention difficulties.

Fourth, the finding that externalizing behaviors demonstrate the strongest discriminant validity has practical implications for clinical assessment.Clinicians should specifically inquire about hyperactive-impulsive behaviors,oppositional tendencies,and conduct problems when evaluating children with epilepsy for possible ADHD.The DBD rating scale,which specifically assesses disruptive behavior disorders, may be particularly useful as a supplementary measure in clinical evaluation.The relatively poor discriminant validity of internalizing measures suggests that anxiety and depression symptoms,while important to assess and treat,are not useful indicators of comorbid ADHD in this population.

Finally,our results concerning risk factors could help develop a stratified approach to screening intensity.Children with identified risk factors should probably have more frequent and/or intensive screening protocols than standard annual screening(according to the ILAE),and this risk stratification would also allow resources to be better targeted on high-risk children.Careful examination for ADHD symptoms is needed in children with refractory seizures,in those who developed epilepsy at young ages,and in children under polytherapy and/or receiving older antiepileptic drugs.

4.4. Limitations and future Directions

However,some limitations should be noted when drawing conclusions from this review.Firstly,the number of studies included in this review was relatively small due to low availability in literature of studies evaluating the validity of screening instruments for ADHD among epilepsy populations.Therefore,it was not possible to perform a *meta*-analysis to pool diagnostic accuracy parameters across studies,nor to carry out subgroup analyses according to epilepsy type,age range,etc.

Secondly,for Layer 2 discriminant validity analysis,we depend largely on one single large,high-quality study.Although this study collects comprehensive data from many different behaviors under the same robust methodology,replication in other diverse populations and contexts would further verify its generalizability;using effect sizes from one single study could limit our inference about whether such discriminant validity patterns can be replicated across different clinical contexts/populations.Nonetheless,leave-one-out sensitivity analysis shows that the pooled estimate is robust and not unduly influenced by any single effect size(see Fig. 6).

Thirdly,included studies differed from one another with respect to how they defined"ADHD".All studies used a clinical evaluation(not just a screening tool),but they might have had different diagnoses according to DSM-IV versus DSM-5,or different types of assessments,or different levels of expertise among clinicians performing the evaluations.This was particularly true when comparing studies done before the revision of the

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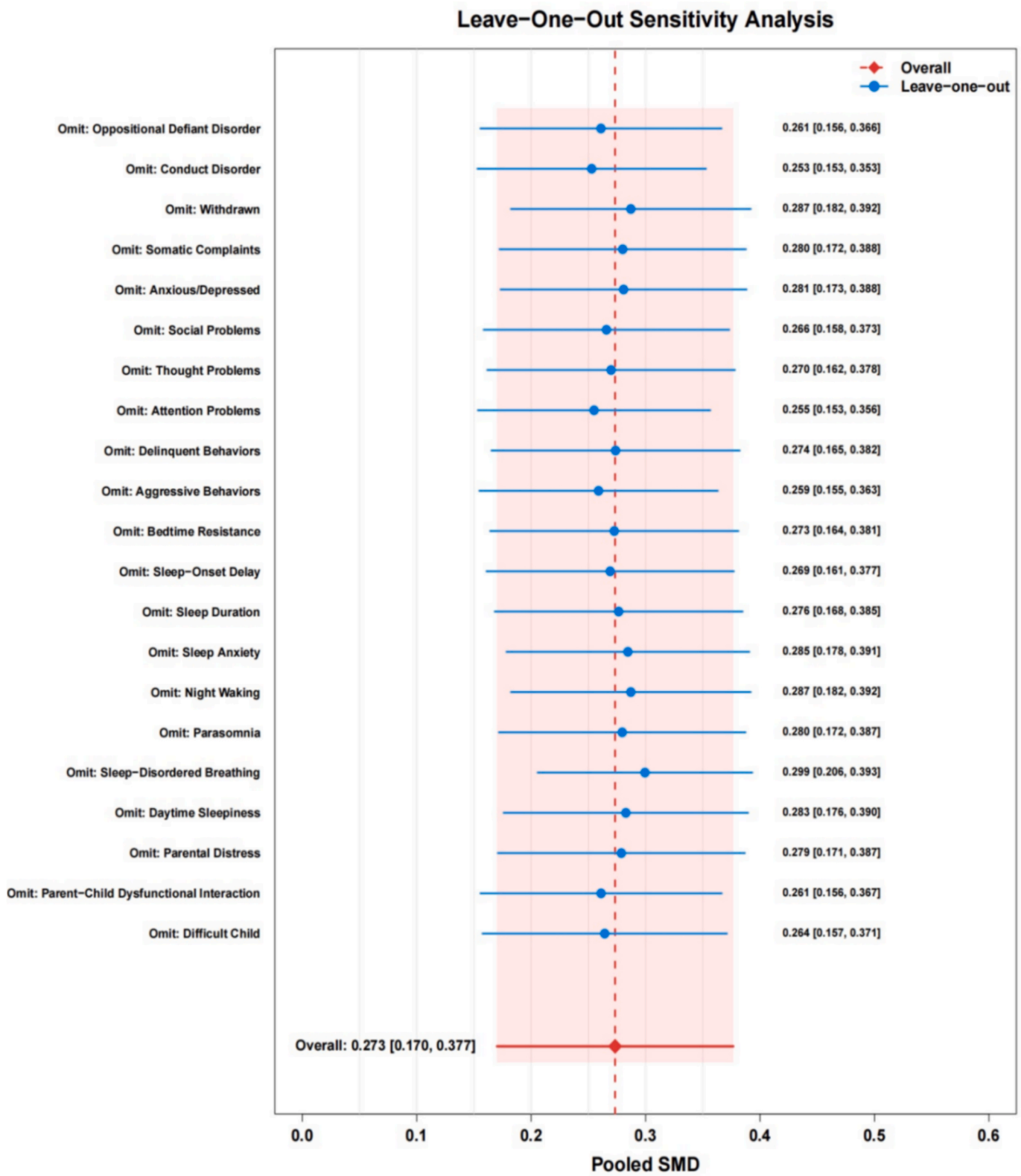


Fig. 6. Leave-one-out sensitivity analysis for discriminant validity *meta*-analysis.Each row shows the pooled effect size(Hedges'g with 95%confidence interval)when that specific measure is excluded from the analysis.The red diamond represents the overall pooled estimate($g = 0.273$,95%CI:0.170–0.377).The shaded area indicates the 95%confidence interval of the overall estimate.Results demonstrate that no single effect size substantially influences the pooled estimate. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

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Table 4
Risk factors associated with comorbid ADHD in children with epilepsy(narrative synthesis).

Study	Risk factor	Definition/Comparison	Effect(95%CI)	p value	Adjusted	Covariates
Alfstad 2016	Early seizure onset(Age at epilepsy onset)	Per year increase in age at onset(continuous variable)	OR 0.86 (0.77–0.96)	0.01	Yes	Mutually adjusted(executive dysfunction,gender, age at onset entered into the final model via step-down procedure)
Alfstad 2016	Executive dysfunction	Executive composite T-score ≤ 35 vs. > 35(1.5 SD below mean)	OR 8.20 (1.80–37.20)	0.006	Yes	Mutually adjusted(executive dysfunction,gender, age at onset entered into the final model via step-down procedure)
Alfstad 2016	Male gender	Male vs.Female	OR 2.90 (1.20–7.30)	0.02	Yes	Mutually adjusted(executive dysfunction,gender, age at onset entered into the final model via step-down procedure)
Karen L. Kwong,2016	Age at seizure onset(per year increase)	Continuous variable(years)	OR 0.70 (0.58–0.83)	<0.0001	No	Not Applicable
Karen L. Kwong,2016	Medical comorbidities	Presence vs.Absence of medical comorbidities	OR 12.82 (4.44–37.03)	<0.0001	No	Not Applicable
Karen L. Kwong,2016	Medical comorbidities	Presence vs.Absence of medical comorbidities	OR 11.23 (3.73–34.48)	<0.0001	No	Not Applicable
Karen L. Kwong,2016	Medical comorbidities	Presence vs.Absence of medical comorbidities	aOR 8.00 (1.73–37.00)	0.008	No	Age at seizure onset
Karen L. Kwong,2016	Number of antiepileptic drugs at time of study	Not explicitly defined as categorical in multivariate model.	aOR 6.22 (1.26–30.64)	0.025	No	Medical comorbidities,Age at seizure onset
Reilly 2014	Developmental Coordination Disorder (DCD)	DCD diagnosis present vs. absent	aOR 4.52 (1.32–15.46)	0.02	Yes	Not fully specified;analysis included epilepsy-related factors and other comorbidities.

Note:For Table 4,N_total refers to the epilepsy-group sample on which the risk-factor comparison(ADHD vs non-ADHD within epilepsy)was based;therefore,case-control studies may report a larger total sample size when external controls are included.Extracted association estimates were synthesized narratively due to heterogeneity in risk factor definitions and effect measures.CI = confidence interval;OR = odds ratio;aOR = adjusted odds ratio.

DSM-IV to those done after the revision of the DSM-5(e.g.,due to changes in the age of onset requirements and symptom thresholds).

Fourth,we were also unable to explore screening tool performance according to ADHD subtype.With so many more children with epilepsy being inattentive presentations rather than hyperactive-impulsive or combined subtypes,screening tools could have different performance characteristics for each ADHD subtype(e.g.,inattentive vs.hyperactive-impulsive vs.combined subtypes).More research is needed on subtype-specific diagnostic accuracy so that future clinical assessment protocols can be developed accordingly.

Future research priorities will be:(1)large-scale,prospective validation study of screen tool(s)among different epilepsy groups with sufficient sample size to explore subgroup effects;(2)comparison-with-comparison study design based on uniform methods for different screen measures;(3)develop new screens specific to epilepsy considering special diagnostic issues of these children;(4)investigate multisource screen procedures taking ratings from parents,teachers,and clinicians together;(5)conduct long-term observation to check whether the current screen can correctly forecast later impairment;(6)implement study on how best to incorporate regular ADHD screening as part of routine health surveillance of young epilepsy patients.

But also,are screening tools useful in certain epilepsies?And if so, which ones?What is the best way to administer them—in a new diagnosis,or a long-standing epilepsy?Are they cost-effective,and what are their effects on patients'quality of life?How can we evaluate whether they are valid for use in different epilepsy subtypes?[38].

5. Conclusions

First,this is the first systematic review/*meta*-analysis showing that the SDQ-Hyperactivity subscale has a good diagnostic accuracy to detect ADHD in children with epilepsy,thus confirming current ILAE recommendations on this screening tool.Second,further externalizing behavior measures(mainly the DBD rating scale)with excellent discriminant validity might be evaluated in order to update clinical assessment protocols.Third,a narrative synthesis of risk factor evidence (Table 4) showed that earlier seizure onset,neurodevelopmental/cognitive vulnerabilities,male sex,medical comorbidities,and higher antiepileptic drug burden might correspond to higher-risk subgroups for whom targeted screening should be performed.

These results have important clinical implications:healthcare providers caring for children with epilepsy should conduct systematic screening using validated instruments starting at diagnosis,or at the age of 6 years,and repeating annually;they should consider more frequent screening if risk factors are present;treatment should be started when ADHD is detected(methylphenidate seems safe and effective);and antiepileptic drug regimens should be adjusted to avoid cognitive side effects and maintain seizure control.

In other words,we can't afford to wait on screening for ADHD with validated instruments in high-risk populations until there is more data regarding stimulant safety in epilepsy.If we did wait,would we miss the chance to identify and intervene sooner in kids with epilepsy and ADHD?Given the high prevalence of this comorbidity,and its negative impact on developmental trajectories,quality of life,and long-term outcomes,this is a clinical and research imperative.

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CRediT authorship contribution statement

Guang Ni: Writing – original draft, Project administration, Methodology, Formal analysis, Data curation. **Dong Meng:** Writing – review & editing, Validation, Supervision, Software, Resources, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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